

NCNS 兼容与适配指南（英汉对照版）

霜泽图书馆汉化组 译

秋起. 乐 MangoGong Temeraire1107 HELPMEEADICE 参与翻译

秋起. 乐 编辑



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新版本译名对照表

原文	译文	原文	译文
Collections	集合	Script constant	脚本常量
Doctrines	学说	Modder	Modder
Folder	Folder	Fleet in being	存在舰队
Grand Doctrines	主要学说	grand battleplan	大规模作战计划
Track	路线	XP	经验
Milestone	里程碑	Arrays	数组
Subdoctrine	子学说	Checkpoint	节点
Mastery	精通度	Hardcoded	硬编码
Reward	奖励	Powered Factories	工厂动力
Factions	阵营	Resources	资源
Rule	规则	SP/Repair Ship	维修舰
Goal	目标	Modules	模块
Manifest	宣言	Defines	定义
Fleets	舰队	World Tension	全球紧张度
Taskforces	特遣队		
Objectives	任务		
Energy	能源		
Coal	煤		
Naval Dominance	制海权		
Naval Supremacy	海军霸权		
Strategic Locations	战略地点		
State	地区		
Bookmark	Bookmark		
GUIfication	图形界面化		
Trigger	条件/触发器		
Modifier	修正		
Effect	效果		

NCNS 兼容与适配指南（中文版）



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欢迎来到《钢铁雄心 IV》最具变革性的一次更新。尽管本次更新并没有像角色系统如此庞大，但却存在一个第二名（集合-Collections）以及非常多的内容。因此我认为这是模组制作者们开始制作模组以来最重要的一次更新。我将假定你的目标是像《Equestria at War》那样（全面整合所有新的特色内容），而不是仅仅完成最低限度的适配让模组能够启动，却忽略了所有炫酷的新功能。我将这些特性分为**玩家内容**（玩家可以直接互动）和**Modder 内容**（玩家不会直接互动）。

DLC 说明

每个 DLC 都包含免费内容和付费内容，NCNS 也不例外。不幸的是，关于哪些内容属于 DLC、哪些不属于 DLC 的信息非常难找。我会尽力做出最优推测，这在某些情况下是正确的——但其最终决定权在 P 社手中。已知属于“DLC”的内容会在下方的子标题中标记为“DLC”。

我不认为那些包含 DLC 元素的内容（例如学说重做）是 DLC 内容。它们只是有一些 DLC 锁定的元素。只有当整个内容都被锁定时，该内容才被视为 DLC 内容。

集合（Collections）：主要的 Modder 内容

集合（Collections）的重要性几乎和角色（Characters）系统相当。这一次，相比大修诸如领导人，将领，顾问之类的东西，集合在很多情况下是数组（arrays）的优异替代。

简而言之，集合（Collections）允许你拥有自动更新的乱序数组（arrays）。你可以在诸如条件（Triggers）、效果（effect）之类的地方用上它。就像脚本常量（Script constant），他们需要有一个前缀，在使用集合的情况下，你可以直接写：“**Collections:all_countries**”，而不是“**all_countries**”。不管怎样，他们十分强大，特别是当你能够让他们相互传递的情况下，甚至是将他们与脚本常量（Script constant）混合使用的情况下。你可以看一下接下来的四个例子（前两个是原版的，后两个是我写的）

```
controllable_balkan_states = {
    input = constant:state_groups.balkans
    name = COLLECTION_FACTION_BALKAN_STATES
}

states_controlled_by_faction_member_and_core_holder = {
    input = game:scope
    operators = {
        faction_members
        controlled_states
        limit = {
            is_core_of = controller
        }
    }
    name = COLLECTION_STATES_CONTROLLED_BY_FACTION_MEMBER_AND_CORE_HOLDER
}

non_capitulated_majors = {
    input = game:all_countries
    operators = {
        limit = {
            is_major = yes
            has_capitulated = no
        }
    }
    name = COLLECTION_NON_CAPITULATED_MAJORS
}

non_capitulated_majors_with_smaller_navy_than_HIP = {
    input = collection:non_capitulated_majors
    operators = {
```



```

limit = {
    naval_strength_comparison = {
        other = HIP
        ratio < 1.0
        sub_unit_def_weights = { # 航母真的让我们很害怕，我们不太关心小型船
只
            carrier = 1.45
            battleship = 1.15
            battle_cruiser = 1
            heavy_cruiser = 0.85
            light_cruiser = 0.7
            destroyer = 0.55
            submarine = 0.3
        }
    }
}
name = COLLECTION_NON_CAPITULATED_MAJORS_WITH_SMALLER_NAVY_THAN_HIP
}

```

集合文件位于 `common/collections/` 文件夹。当然，类似于临时变量，如果你在其他地方不需要他们，也可以“匿名”定义。

我发现了下列两个未登记的，与集合（collections）相关的条件（trigger）：
“`count_in_collection`”和“`has_resources_in_collection`”

学说(Doctrines): 主要的玩家内容

大多数学说都被重置了，但是还有一小部分还能看到旧版的一些影子（比如特种作战学说，special forces doctrines）。学说几个核心概念：**folder**，**主要学说（grand doctrine）**，**里程碑（milestone）**，**子学说（subdoctrine）**，**精通度（mastery）**，**与奖励（reward）**。这里的所有内容都在 `common/doctrines/` 路径下。它的修改性非常强，而且文档也很完善！

Folder

Folder 是最高级的学说元素。在默认的情况下有三个，即海陆空，但是您也可以增加更多。在军种一下，您可以选择一个**主要学说（grand doctrine）**。

```
land = {
```

```

allowed = {
    always = yes
}
name = "land_doctrine_folder"
ledger = army
ledger_gfx = GFX_land_doctrine_folder_icon
tab_gfx = GFX_landdoctrine_tab_large
color_frame = 1
sound = ui_doctrine_tab_land
}

```

主要学说

主要学说 (grand doctrines) 应该对咱们来说很熟悉：他们就是诸如“存在舰队” (fleet in being) 和“大规模作战计划” (grand battleplan) 的东西。他们在刚被选择的时候有奖励 (bonus)，然后在点完了一条**路线** (原文这里是 **Track**) 之后也有奖励。通常解锁一个主要学说要花费 100 对应的军种的经验 (XP)。

它们的语法一开始看起来有些奇怪，但是你会习惯的（步兵里程碑之类的东西是用作成就的）：

```

grand_battleplan = {
    folder = land

    name = GRAND_DOCTRINE_GRAND_BATTLEPLAN
    description = GRAND_DOCTRINE_GRAND_BATTLEPLAN_DESC
    icon = GFX_doctrine_grand_battleplan_medium
    available = {
        always = yes
    }

    xp_cost = 100
    xp_type = army

    ai_will_do = {
        base = 2
    }

    tracks = {
        infantry
        combat_support
        armor
    }
}

```

```

    operations
}

# EFFECTS
enable_tactic = tactic_planned_attack

max_planning = 0.1
command_power_gain = 0.25

milestones = {
    {
        #Infantry
        additional_brigade_column_size = 1
        enable_tactic = tactic_elastic_defense
        max_dig_in_factor = 0.05

        effect = {
            if = {
                limit = {
                    NOT = {
                        has_variable = infantry_milestone_var
                    }
                }
                set_variable = { infantry_milestone_var = 1 }
            }
            else = {
                add_to_variable = { infantry_milestone_var = 1 }
            }
        }
    }
}

{
    #Artillery/Support
    coordination_bonus = 0.1
    enable_tactic = tactic_overwhelming_fire

    effect = {
        if = {
            limit = {
                NOT = {

```

```

        has_variable = support_milestone_var
    }
}
set_variable = { support_milestone_var = 1 }
}
else = {
    add_to_variable = { support_milestone_var = 1 }
}
}
}
{
    #Armor
    land_reinforce_rate = 0.02
    enable_tactic = tactic_breakthrough
    effect = {
        if = {
            limit = {
                NOT = {
                    has_variable = armor_milestone_var
                }
            }
            set_variable = { armor_milestone_var = 1 }
        }
        else = {
            add_to_variable = { armor_milestone_var = 1 }
        }
    }
}
}
{
    #Operations
    coordination_bonus = 0.1
    max_command_power = 30

    effect = {
        if = {
            limit = {
                NOT = {
                    has_variable = operations_milestone_var

```


子学说（Subdoctrine）

每条路线可以消耗一定 XP（通常为 100）选择一项子学说，并可自定义精通条件。选定该子学说后，您将立即获得一项加成，随后根据精通度逐步推进。路径上设有多项奖励，每项奖励都有独特增益效果。

```
air_cavalry = {
    track = combat_support
    name = SUBDOCTRINE_AIR_CAVALRY
    description = SUBDOCTRINE_AIR_CAVALRY_DESC
    icon = GFX_doctrine_air_cavalry_medium

    xp_cost = 100
    xp_type = army

    available = {
        is_special_project_completed = sp:sp_air_helicopter
    }
    visible = {
        has_dlc = "Gotterdammerung"
    }

    ai_will_do = { base = 0 }

    # EFFECTS
    effect = {
        add_tech_bonus = {
            bonus = 0.5
            uses = 2
            category = helicopter_tech
            name = SUBDOCTRINE_AIR_CAVALRY
        }
    }

    rewards = {
        airborne_observers = {
            helicopter_recon = {
                battalion_mult = {
                    category = category_artillery
                    soft_attack = 0.1
                }
            }
        }
    }
}
```

```

        }
        max_organisation = 10
    }
}
service_and_support = {
    category_helicopter_support_companies = {
        max_strength = 0.03
    }
}
rescue_liasions = {
    helicopter_field_hospital = {
        battalion_mult = {
            category = category_all_infantry
            max_strength = 0.05
            default_morale = 0.05
        }
        max_organisation = 5
        casualty_trickleback = 0.05
    }
}
transport_assault = {
    helicopter_transport = {
        battalion_mult = {
            category = category_light_infantry
            maximum_speed = 0.2
            breakthrough = 0.05
        }
        breakthrough = 0.05
        max_organisation = 10
    }
}
intervention_units = {
    helicopter_brigade = {
        battalion_mult = {
            category = category_all_infantry
            soft_attack = 0.1
        }
        max_organisation = 10
    }
}

```

```
}  
}  
}  
}
```

精通度（Mastery）

精通度是一项新的资源，可以通过训练或使用特定单位进行战斗而获得。与 XP 相比，这种资源的划分更为精细，且（至少在当前阶段）仅用于学说体系。某些东西可以修正精通度的获取效率，而这一点，正好可以用来强化旧的学说增益效果！请看下面的例子：

```
add_mastery_bonus = {  
    bonus = 0.2  
    days = 180  
    grand_doctrine = mass_assault  
}
```

奖励（Reward）

奖励是子学说中特定**精通度**等级的“节点”，绝大多数增益效果都来源于此。

阵营（Factions）：主要的玩家 DLC 内容

现在阵营系统已突破外交触发器的限制，支持高度自定义了。系统提供了丰富的可配置选项，且相关文档极为详尽，强烈建议先阅读文档！需要注意的是：虽然可以在无 DLC 环境中使用“create_faction_from_template”效果，但所有高级阵营功能均需拥有 DLC 方可使用。

回来了？好的，我们来看一个阵营的例子。看起来很简单，但其中隐藏着很多东西：

```
faction_template_river_coalition = {  
    name = river_coalition_faction  
    manifest = faction_manifest_ruling_the_rivers  
  
    goals = {  
        faction_goal_an_infantry_army # short  
        faction_goal_a_great_fortress # medium  
        faction_goal_defeat_the_empire # long  
    }  
  
    default_rules = {
```

```

        call_to_war_rule_faction_leader_offensive_only
        faction_peace_take_claimed_focus
        government_in_exile_allowed
    }

    icon = GFX_faction_icon_allies
}

```

规则（Rule）

规则均在 `common/rules/` 路径中定义，并分组到 `common/rules/groups`。下面是其中之一：

```

all_to_war_rule_faction_leader_offensive_only = {
    type = call_to_war_rules
    trigger = {
        OR = {
            is_faction_leader = yes
            has_defensive_war = yes
        }
    }

    visible = {
        has_faction_template = faction_template_river_coalition
    }

    modifier = {
        defensive_war_stability_factor = 0.05
    }
}

```

目标（Goal）

以下是目标的示例。请注意，目标可以是短期、中期或长期目标：

```

faction_goal_defeat_the_empire = {
    name = defeat_the_empire_name
    description = defeat_the_empire_desc
    category = long_term
    group = FOCUS_FILTER_ANNEXATION

    allowed = {
        is_griffon_continent_country = yes
    }
}

```

```

NOT = {
    original_tag = GRI
    original_tag = BRZ
    original_tag = FEA
    original_tag = ANG
    original_tag = KAT
    original_tag = PYT
    original_tag = ROU
    original_tag = YAL
    original_tag = STW
}

}

completed = {

    GRD = {
        OR = {
            is_in_faction_with = ROOT
            has_capitulated = yes
            exists = no
            is_subject_of = ROOT
        }
    }

}

complete_effect = {
    add_faction_power_projection = 500
    custom_effect_tooltip = generic_skip_one_line_tt
    custom_effect_tooltip = THE_FACTION_LEADER
    add_faction_initiative = 1
    custom_effect_tooltip = generic_skip_one_line_tt
    every_faction_member = {
        tooltip = INFLUENCE_TOP_THREE
        visible_when_empty = yes
        limit = {
            faction_influence_rank < 4
        }
        add_political_power = 100
    }
}

```



```
}  
}
```

宣言（Manifest）

这就是宣言的样子：

```
faction_manifest_the_greatest_navy_in_the_world = {  
    name = faction_manifest_the_greatest_navy_in_the_world_name  
    description = faction_manifest_the_greatest_navy_in_the_world_desc  
    is_manifest = yes  
    ratio_progress = {  
        total_amount_collection = collection:non_capitulated_majors  
        completed_amount_collection =  
collection:non_capitulated_majors_with_smaller_navy_than_HIP  
  
        scale = {  
            navy_fuel_consumption_factor = -0.25  
            fuel_gain_factor = 0.25  
        }  
        progress_sections = {  
            manifest_fulfilled = {  
                min = @manifest_fulfilled_value  
                modifier = {  
                    production_lack_of_resource_penalty_factor = -0.2  
                }  
            }  
        }  
    }  
}
```

如你所见，内容非常丰富！查阅文档和示例，以便更好地了解系统。此外，还有一些成员升级选项，例如人力共享，但这些选项不太常被修改（至少就我的经验而言）。

AI 阵营威胁：Modder 内容

随着新的阵营功能升级，我们也能够创建威胁，并能够告诉 AI 建立哪些威胁。这里有一个例子：

```
alicorn_sunset_north = { # northern front down to crystal ponies
```

```
name = theater_alicorn_sunset_north
regions = {
    84 # Featherfall Dale
    73 # Yaket Range
    74 # Stonefields
    71 # Crystal Heart
    72 # Crystal Range
    65 # Severyana
    102 # Central Yakistan
    103 # Greater Yakistan Valley
    12 # Frozen North
    191 # Bronze Mountains
    121 # Bright Valleys
}

preferred_countries = {
    PLB
}

cancel = {
    NOT = { has_war_with = EQS }
}

ai_will_do = {
    base = 0
    modifier = {
        add = 5
        OR = {
            original_tag = CHN
            is_in_faction_with = CHN
        }
        has_war_with = EQS
    }
    modifier = {
        add = 1
        is_in_faction_with = CHN
        has_war_with = EQS
    }
}
```

```
}  
}
```

AI 海军：Modder 内容

AI 海军终于可以进行深度自定义了！除了装备之外，我们还可以告诉不同的国家（tag）要组建什么样的特遣队和舰队，以及何时组建。我们还可以告诉它们应该发挥什么功能。这些功能都有详细的文档说明，快去阅读吧！

回来了？太好了。这里有四个核心概念：**舰队（Fleets）**、**特遣（舰）队（Taskforces）**、**目标（Goals）**和**任务（Objectives）**。目标是高层级的，例如保护贸易路线；任务则是低层级的，例如保护特定的贸易路线。舰队是由特遣队组成的编队，而特遣队则负责执行特定的任务。任务由人工智能自行设定，因此是**硬编码**（意思就是不能修改）的。不过，其他三个概念都是可以修改的！

舰队（Fleets）

以下是一个示例舰队文件：

```
HIP_dominance_fleet_1 = {  
    required_taskforces = {  
        HIP_StrikeForce_1 = 1  
        HIP_PatrolDominanceForce_1 = 3  
        HIP_PatrolReconForce_1 = 3  
    }  
    optional_taskforces = {  
        HIP_StrikeForce_1 = 1  
        HIP_PatrolDominanceForce_1 = 3  
        HIP_PatrolReconForce_1 = 3  
    }  
}  
  
HIP_raiding_fleet_1 = {  
    required_taskforces = {  
        HIP_ConvoyRaiding_1 = 4  
    }  
    optional_taskforces = {  
        HIP_ConvoyRaiding_1 = 2  
    }  
}
```

```

HIP_escort_fleet_1 = {
    required_taskforces = {
        HIP_ConvoyEscort_1 = 2
    }
    optional_taskforces = {
        HIP_ConvoyEscort_1 = 4
    }
}

HIP_minelaying_fleet_4 = {
    required_taskforces = {
        HIP_MineLaying_1 = 1
    }
}

```

每支舰队都包含若干**特遣队**。有些舰队只有一个特遣队（例如布雷舰队），而有些舰队则有多个特遣队（例如主力舰队）。让我们来看看一支 HIP（示例 TGA）主力舰队的组成：

特遣队（Taskforces）

```

HIP_StrikeForce_1 = {
    allowed = {
        original_tag = HIP
    }
    ai_will_do = {
        factor = 1
    }
    mission = { naval_strike }
    min_composition = {
        destroyer = {
            amount = 8
        }
    }

    optimal_composition = {
        carrier = {
            amount = 2
        }
        battleship = {
            amount = 4
        }
    }
}

```

```

    }
    heavy_cruiser = {
        amount = 4
    }
    light_cruiser = {
        amount = 4
    }
    destroyer = {
        amount = 36
    }
}

HIP_PatrolReconForce_1 = {
    allowed = {
        original_tag = HIP
    }
    ai_will_do = {
        factor = 5
    }
    mission = { naval_patrol }
    min_composition = {
        light_cruiser = {
            amount = 2
        }
    }
    optimal_composition = {
        light_cruiser = {
            amount = 3
        }
    }
}

HIP_PatrolDominanceForce_1 = {
    allowed = {
        original_tag = HIP
    }
    ai_will_do = {

```

```

        factor = 5
    }
    mission = { naval_patrol }
    min_composition = {
        destroyer = {
            amount = 4
        }
    }

    optimal_composition = {
        heavy_cruiser = {
            amount = 3
        }
        destroyer = {
            amount = 12
        }
    }
}

```

如您所见，舰队列表中列出了实际舰船的数量及其用途！然后，这些信息会按如下方式输入到目标系统中（以 HIP 为例）：

目标（Goals）

```

HIP_naval_invasion_support = {
    objective_type = naval_invasion_support

    available_for = {
        HIP
    }

    min_priority = 10
    max_priority = 20
}

HIP_mine_sweeping = {
    objective_type = mines_sweeping

    available_for = {
        HIP
    }
}

```

```
}

    min_priority = 3
    max_priority = 8
}

HIP_invasion_defense = {
    objective_type = naval_invasion_defense

    available_for = {
        HIP
    }

    min_priority = 10
    max_priority = 18
}

HIP_coast_defense = {
    objective_type = coast_defense

    available_for = {
        HIP
    }

    min_priority = 1
    max_priority = 10
}

HIP_convoy_protection = {
    objective_type = convoy_protection

    available_for = {
        HIP
    }

    min_priority = 1
    max_priority = 5
}
```

```
HIP_convoy_raiding = {  
    objective_type = convoy_raiding  
  
    available_for = {  
        HIP  
    }  
  
    min_priority = 3  
    max_priority = 7  
}
```

```
HIP_naval_dominance = {  
    objective_type = naval_dominance  
  
    available_for = {  
        HIP  
    }  
  
    min_priority = 1  
    max_priority = 13  
}
```

```
HIP_mine_laying = {  
    objective_type = mines_planting  
  
    available_for = {  
        HIP  
    }  
  
    min_priority = 1  
    max_priority = 5  
}
```

```
HIP_training = {  
    objective_type = training  
  
    available_for = {  
        HIP  
    }  
}
```



```

min_priority = 10
max_priority = 20
}

```

这样一来，AI 的行为就被“限制”了，例如，HIP 从不全力布雷，而是一直致力于海岸防御。

国策动态时长与叠加图层：Modder 内容

历经多年，它终于来了！这意味着您可以随意调整国策时长，并在国策图中实时更新。以下是如何将国策时长缩短 70 天的方法：

```

reduce_focus_completion_cost = {
    focus = PHI_national_defense_act
    cost = 70
}

```

国策也支持叠加图层层，即显示在国策上方的元素，类似于标题栏样式或联合国策的标志（但可以与它们结合使用）。它们只需向国策代码添加一个如下所示的参数：

```

overlay = GFX_focus_fast_overlay_generic_clock

```

能源和煤（Energy & Coal）：玩家内容

游戏中新增了一种资源——煤炭，以及“工厂动力”的概念。此外，还有一些只有在工厂完全运转时才会生效的加成效果，不过这些效果相当罕见。

```

production_speed_buildings_powered_factor, industrial_capacity_factory_powered,
industrial_capacity_dockyard_powered

```

<译>已供能的建造速度，已供能的工厂产出，已供能的船坞产出

```

factory_energy_consumption, local_factory_energy_consumption,
local_resources_<resource>_factor

```

<译>工厂能源消耗，当地工厂能源消耗，当地资源<resource>获取效率

请注意，**经济活动越活跃，消耗的能源就越多**。目前，原版游戏没有增加能源获取或直接补充能源的科技，也没有添加这些科技所需的修正。希望以后能有！（所以为什么不能用石油供能）

此外，请注意，新增了一个 NAI 定义：

```

NDefines.NAI.MIN_FACTORIES_TO_WANT_TO_IMPORT = {  --AI 必须拥有的最小民用工厂数量，
才能考虑进口某种战略资源。默认值为 0，数组应随着新资源的出现或顺序的改变而更新。

```

```

    10, -- oil -- #由于社会陈旧，欧洲战乱加剧
    0, -- aluminium
    0, -- rubber
    0, -- tungsten

```

```
0, -- steel
0, -- chromium
10, -- coal
0, -- crystals
}
```

任何添加了新的 resources 的模组必须更新此代码，否则 AI 无法理解该资源如何运作！

制海权（Naval Dominance）：玩家内容

海军霸权（Naval Supremacy）已不复存在，取而代之的是**制海权**。整个系统运作方式截然不同：海域地形会影响达到制海权所需的数值，不同的任务会以不同的方式与制海权互动，舰船的制海权数值也千差万别（但可以在设计器中查看！）。这是一项**重大变革**，所以请深入探索，看看其中究竟有何变化！尤其需要注意的是，`common/00_terrain.txt` 文件需要制海权才能正常工作。

游骑兵特种部队：玩家内容

我们终于拥有了专为森林和丛林而组建的特种部队，以及相应的特种部队理论（如果您还有以力御暴 DLC）。

战略地点（Strategic Locations）：玩家内容

战略地点是一项新功能，允许我们自定义特定省份或州内建筑的最大数量。它们分为两类：省和州。令人惊喜的是，它们的定义方式相同，只是效果不同，正如冲绳州的文件（在 history 目录下）所示：

```
strategic_province_location = {
    defensible_coastline = 10124
}
strategic_state_location = {
    favorable_approach = 1123
}
strategic_province_location = {
    defensive_chokepoint = 11932
}
```

他们定义在 `common/strategic_locations/` 目录下，可以容纳任何建筑类型（有数量上限），他们在游戏中添加的效果相同，但请注意，它必须在 **State** 作用域内！如下：

```
defensible_coastline = {
    coastal_bunker = 2
}
```

```
favorable_approach = {  
    air_base = 2  
}
```

玩家扩展内容

以下是玩家会关心的一些更新和新增内容，但这些内容并不属于全新的类别。

新增 DLC 科技：维修舰（SPs）和新舰船模块（Modules）

海军不再仅仅是战斗舰。**维修舰**和**支援舰**能为你的海军提供强大的增益效果，甚至能“拯救”那些原本会被击沉的舰船！它们通过两种新的装备类型和一个静态修正代码来实现：

```
# This is scaled based on support ratio  
naval_general_support = {  
    max_organisation = 0.2  
    navy_max_range_factor = 0.2  
    naval_morale_factor = 0.2  
}  
  
# This is scaled based on support ratio  
naval_repair_support = {  
    naval_critical_effect_factor = -0.25  
    naval_ship_recovery_chance = 0.1  
    naval_attrition = -0.075  
}
```

新增了**先进声呐（Sonar 3）**和**现代声呐（Sonar 4）**以及**水上飞机母舰（Seaplane tenders）**等新型装备（再也不用担心隐形潜艇了）。此外，我们还以特别项目的形式，增建了**潜艇航母**和**护航航母**，这些项目都非常出色。（潜艇航母你确定出色吗.....）。

航母过度折叠机制重构

终于！旧式航母作战方式的定义被移除了。现在，作战方式基于现役舰艇的数量，因此战斗规模越大，能够有效作战的飞机就越多：

```
NAVAL_COMBAT_PLANE_MIN_STACKING_PENALTY = 80, --  
引入惩罚之前，海战中飞行的飞机数量。  
  
NAVAL_COMBAT_PLANE_STACKING_PENALTY_EFFECT = 0.025, -- 每架飞机数  
量超过最优值，可起飞飞机的数量就会减少 0.025%。受收益递减规律约束。  
  
SHIP_SILHOUETTE_VALUE_PLANES_CAPITAL = 10, -- 对于动态平  
面功效，轮廓值（名义上是平面，但非常抽象）
```

```

SHIP_SILHOUETTE_VALUE_PLANES_SCREEN = 5,           -- 同上，如果
能按照类型拆分就更好了，但这会有问题。

SHIP_SILHOUETTE_VALUE_PLANES_CARRIER = 16,         -- 同上
SHIP_SILHOUETTE_VALUE_PLANES_SUPPORT = 3,           -- 同上
SHIP_SILHOUETTE_VALUE_PLANES_CONVOY = 4,            -- 同上
SHIP_SILHOUETTE_VALUE_PLANES_SUBMARINE = 7,          -- 同上

SCREEN_CAP_REDUCTION_FACTOR = 0.02,                 -- 如果存在屏幕
盖，则减少屏幕轮廓权重，屏幕值 * 1/(1+屏幕盖*权重)

```

由此可见，面对一支由 1 艘航母、1 艘战列舰、1 艘重巡洋舰、2 艘轻巡洋舰和 8 艘驱逐舰组成的‘小型’特遣舰队，能够无惩罚起飞的飞机数量为 $80+16+10+10+(52+58)(1/(1+30.02))=163.2$ 。

世界紧张度大改

世界紧张度的定义发全球紧张度在 defines（定义文件）中发生了一些重大变化。举例来说，以前建立傀儡国会产生紧张度，现在完全不会了。**TENSION_STATE_VALUE** 过去通常会被除以 20，但现在不再这么做了；而且 **TENSION_PEACE_FACTOR** 现在会受到一个新参数 **TENSION_PEACE_THREAT_FACTOR** 的削减。无理由宣战变得更加严酷，吞并核心领土现在会降低世界紧张度，强制更迭政权现在也会降低世界紧张度。夺取主力舰不再产生紧张度，索取战争赔款、资源权、拆除工业或投降也不再产生紧张度。

基于舰载机数量的对舰伤害

这真是个很棒的设定。现在，航母上的飞机越多，它就越擅长猎杀敌舰。不同的航母模块支持的飞机数量不同，所以它有很大的改装空间！

```

add_stats = {
    carrier_size = 2
    build_cost_ic = 1500
    carrier_surface_detection = 5
    carrier_sub_detection = 2
}

```

新建筑

我们新增了海军补给枢纽（Naval supply hubs）和海军总部（naval HQs）。海军补给枢纽可以降低舰队的补给消耗，而海军总部允许你指派海军上将以提供大范围的加成（仅限 DLC）。有趣的是，这些新建筑要求该省份至少拥有 8 级港口——我以前以为这是不可能实现的。具体语法如下：

```
building_prerequisite = {
    building_dependency = {
        building = naval_base
        level = 8
    }
}
```

地区（State）类别调整

现在，各地区（State）类别可以支持限制其关联的最高建筑等级——例如

```
buildings_max_level = {
    naval_base = 6
    air_base = 4
}
```

目前，这些仅在岛屿上受到上限限制。此外，原版新增了一个地区类别——**大型岛屿**（large island）。该类别拥有 3 个建筑槽位，并且海军基地和空军机场也受到限制。

杂项改动

这里要列出的内容实在太多了，所以建议大家自己去翻翻文件。举个例子，现在的冰雪严寒天气更为严酷（这下苏联更难推了），登陆作战的机制也完全变了。中型火炮科技已经不存在了，相关模块被整合进了轻型和重型火炮中。

Modder 扩展内容

以下是一些玩家可能不会注意到的更新和新增内容，它们并非全新的类别。

新的 Bookmark 内容

Bookmarks 现在支持一些新的标注方式。在国家选项下，Bookmarks 提供以下两种选项：

```
new_content = yes
override_leader_portrait = JAP_emperor_hirohito
```

这样，你就可以让图标闪烁来通知玩家有新内容，还可以显示与实际不同的头像！

可点击的核心圈窗口按钮和核心圈移动

原版的图形界面化（GUIfication）进程仍在继续！现在，国策内嵌窗口已支持带有效果和触发器的按钮。例如：

```

    army_faction_stamp_of_approval = {
        available = {
            has_political_power > 74
            JAP_army_faction_is_not_dominant = yes
            IF = {
                limit = {
                    NOT = { has_country_flag =
JAP_interfered_in_imperial_influence_flag }
                }
                NOT = { has_country_flag =
JAP_interfered_in_imperial_influence_flag }
            }
            ELSE_IF = {
                limit = {
                    has_country_flag =
JAP_interfered_in_imperial_influence_flag
                }
                custom_trigger_tooltip = {
                    tooltip =
JAP_imperial_influence_interference_reset_days_tt
                    NOT = { has_country_flag =
JAP_interfered_in_imperial_influence_flag }
                }
            }
            NOT = {
                has_country_flag = JAP_blocked_from_favoring_the_army_flag
            }
        }
        click_effect = {
            IF = {
                limit = {
                    JAP_army_faction_is_tier_1 = yes
                }
                add_political_power = -200
            }
            ELSE = {
                add_political_power = -75
            }
        }
    }

```

```

        custom_effect_tooltip = generic_skip_one_line_tt

        JAP_big_army_faction_gain = yes
        set_country_flag = {
            flag = JAP_interfered_in_imperial_influence_flag
            value = 1
            days = 180
        }
    }
}

```

我们现在还可以设国策内镶嵌窗口图案的偏移量，或者更确切地说是 `override_position`:

```

        override_position = {
            x = 4400
            y = 700
            trigger = {
                has_game_rule = {
                    rule = obsolete_focus_branches_visibility
                    option = HIDE
                }
                has_completed_focus = JAP_sea_purge_the_kodoha_faction
            }
        }

        override_position = {
            x = 6000
            y = 700
            trigger = {
                has_game_rule = {
                    rule = obsolete_focus_branches_visibility
                    option = HIDE
                }
                has_completed_focus = JAP_revere_the_emperor_destroy_the_traitors
            }
        }
    }
}

```

这部分内容应放在 `common/national_focus/` 目录下与该嵌入窗口相关代码的文件。

国家和地区脚本常量

我们为国家和地区新增了脚本常量，以及一个使用如下格式的新触发器：

```
any_country_with_original_tag_of = {
    tooltip = TRIGGER_ANY_COUNTRY_STARTS
    target   = constant:country_groups.literally_china
    OR = {
        has_completed_focus = CHI_ledo_road
        has_completed_focus = CHI_sea_ledo_road
    }
}
```

以下是脚本常量的具体形式：

```
country_groups = {
    schema = {
        any_key = yes
        array = country
    }
    north_zebrica = {
        HIP
        WIN
        BAT
        CTH
        ZAR
        WAR
        TBK
        ZUM
        CVA
        TRO
        HIE
        CDR
        CGI
        BKH
    }
}
```

类似的概念也适用于 state_groups：

```
state_groups = {
    schema = {
        any_key = yes
    }
}
```



```
    array = state
}

# 巴尔干半岛（包括希腊大陆、土耳其和 1936 年罗马尼亚全境，但不包括匈牙利）
balkans = {
    102
    853
    103
    109
    104
    804
    45
    764
    107
    108
    803
    105
    802
    106
    970
    934
    44
    805
    163
    852
    736
    48
    212
    801
    211
    77
    971
    184
    731
    185
    47
    186
    46
    81
```

```
82
83
84
76
79
766
78
80
341
797
}
}
```

强制关联不可通行状态

我们终于不用再面对不可通行地区归属权疯狂‘反复横跳’的问题了。现在的不可通行地区支持 `force_link_ownership_to = [STATE_ID]` 这一参数，它能将其所有权与目标地区的所有者强制绑定。该参数在 history 文件中设置，通常紧跟在 `impassable = yes` 之后。

杂项新条件（Triggers）

<code>has_resources_in_collection</code>	- 国家作用域， 集合相关
<code>is_on_same_continent_as</code>	- 国家，地区（State）作用域
<code>faction_power_projection</code>	- 国家作用域
<code>has_offensive_war_without_friend</code>	- 国家作用域
<code>can_assign_supportive_scientist_to_faction</code>	- 国家作用域
<code>faction_upgrade_level</code>	- 国家作用域
<code>faction_influence_score</code>	- 国家作用域
<code>faction_influence_ratio</code>	- 国家作用域
<code>has_manpower_to_become_leader</code>	- 国家作用域
<code>has_industry_to_become_leader</code>	- 国家作用域
<code>count_in_collection</code>	- 国家作用域， 集合相关
<code>collection_contains</code>	- 任意作用域， 集合相关
<code>has_doctrine</code>	- 国家作用域
<code>has_completed_subdoctrine</code>	- 国家作用域
<code>has_subdoctrine_in_track</code>	- 国家作用域
<code>has_mastery</code>	- 国家作用域
<code>has_completed_track</code>	- 国家作用域
<code>has_tactic</code>	- 国家作用域

has_enemy_naval_control	- 国家作用域
has_naval_control	- 国家作用域
any_state_of	- 任意作用域
any_collection_element	- 任意作用域，集合相关
all_collection_elements	- 任意作用域，集合相关
all_country_of	- 任意作用域
all_claimant	- 地区（State）作用域
any_country_of	- 任意作用域
any_country_with_original_tag_of	- 任意作用域
any_other_country_of	- 任意作用域
any_other_country_with_original_tag_of	- 任意作用域
has_faction_template	- 国家作用域
faction_manifest_fulfillment	- 国家作用域
collection_size	- 任意作用域，集合相关
faction_influence_rank	- 国家作用域
has_completed_faction_goal	- 国家作用域
has_faction_goal	- 国家作用域
faction_goal_fulfillment	- 国家作用域
is_leading_army_in_province	- 角色（Character）作用域
has_active_rule	- 国家作用域
has_faction_research_unlocked	- 国家作用域
has_faction_military_unlocked	- 国家作用域
has_enough_influence_for_leadership	- 国家作用域
compare_ideology_with_faction	- 国家作用域

杂项新修正（Modifier）

army_claim_attack_factor
 naval_critical_score_chance_factor_against
 naval_supply_consumption_factor
 navy_capital_ship_attack_factor_against
 navy_screen_attack_factor_against
 navy_submarine_attack_factor_against
 planning_decay_rate_factor
 air_invasion_division_cap
 air_invasion_preparation
 army_claim_defence_factor
 army_experience_from_volunteers
 army_retreat_speed_factor
 modifier_army_sub_unit_category_rocket_artillery_attack_factor
 rangers_special_forces_contribution_factor

autonomy_manpower_share_from_subjects
resources_to_overlord_factor
<Doctrine>_doctrine_cost_factor
<Doctrine>_mastery_gain_factor
<Doctrine>_track_mastery_gain_factor
<IdeaCategory>_category_type_cost_factor
<IdeaGroup>_cost_factor
<ModifierStat>_factor # "naval_general_support", "naval_repair_support"
<ModifierStatValue>_value_factor # "naval_general_support",
"naval_repair_support"
<Technology>_cost_factor
air_invasion_plan_cap
air_invasion_prep_days
annex_subject_cost_factor
automatic_grant_medal_chance
experience_gain_<Unit>_mission_factor
factory_energy_consumption
industrial_capacity_dockyard_powered
industrial_capacity_factory_powered
local_resources_<Resource>_factor
naval_invasion_plan_cap
naval_invasion_prep_days
out_of_power_impact_factor
production_cost_<Building>_factor
production_speed_buildings_powered_factor
repair_speed_<Building>_factor
shore_bombardment_collateral_damage_factor
special_forces_doctrine_cost_factor
unit_medal_effectiveness
navy_capital_ship_defence_factor_against
navy_screen_defence_factor_against
navy_submarine_defence_factor_against
faction_influence_contribution_factor
faction_influence_war_score_factor
faction_subject_contribution_gain
intelligence_operation_speed
amphibious_invasion_against
experience_gain_<Unit>_combat_factor
experience_gain_<Unit>_training_factor
headquarters_experience_gain_factor
invasion_preparation_against
max_organisation
naval_hit_chance_against
naval_invasion_division_cap
naval_mission_xp_factor

naval_ship_recovery_chance
naval_ship_recovery_chance_factor
spotting_chance_against
local_factory_energy_consumption
can_guarantee_other_ideologies
<ideology>_drift_from_guarantees

已弃用效果

create_faction - use `create\_faction\_from\_template`

其他新效果

reduce_focus_completion_cost	- 国家作用域
strategic_province_location	- 地区 (State) 作用域
strategic_state_location	- 地区 (State) 作用域
select_tech_tree_icon_origin	- 国家作用域
add_faction_power_projection	- 国家作用域
add_state_resistance_compliance_modifier	- 任意作用域?
remove_state_resistance_compliance_modifier	- 任意作用域?
set_faction_member_upgrade_min	- 国家作用域
set_faction_upgrade	- 国家作用域
every_faction_member	- 国家作用域
add_faction_influence_score	- 国家作用域
add_faction_goal_slot	- 国家作用域
add_faction_influence_ratio	- 国家作用域
set_grand_doctrine	- 国家作用域
set_sub_doctrine	- 国家作用域
add_mastery	- 国家作用域
add_daily_mastery	- 国家作用域
unlock_tactic	- 国家作用域
add_mastery_bonus	- 国家作用域
create_faction_from_template	- 国家作用域
set_faction_rule	- 国家作用域
set_faction_manifest	- 国家作用域
add_faction_goal	- 国家作用域
remove_faction_goal	- 国家作用域
add_faction_initiative	- 国家作用域
every_collection_element	- 任意作用域
set_faction_research_unlocked	- 任意作用域

set_faction_military_unlocked

- 任意作用域

新变量

世界资源开采总量

global_resource_extracted_total

全球特定资源的开采总量，资源类型定义在 target global_resource_extracted@steel 中
global_resource_extracted@<resource> global_resource_extracted@< resource>

检查当前国家在阵营中的影响力比例

faction_influence_ratio

该国阵营掠夺的资源总量

faction_resource_extracted_total

该国阵营提取的特定资源总量，资源类型在 target faction_resource_extracted@steel 中定义
faction_resource_extracted@<resource>

检查当前国家阵营投影的实力值

faction_power_projection

检查当前国家在阵营中的影响力

faction_influence_score

检查当前国家阵营宣言的完成情况

faction_manifest_fulfillment

检查在当前国家阵营中的影响力等级

faction_influence_rank

海军总部建筑总数

total_constructed_naval_headquarter

海军补给枢纽建设总量

total_constructed_naval_supply_hub

支援船类型设备总产量

total_equipment_produced_support_ship

NCNS Compatibility and Feature Guide

英文原版

NCNS Compatibility and Feature Guide



Real photograph of PDX creating collections ([image source](#))

Welcome to probably **the most game-changing update in HOI4 history**. While nothing is quite as big as characters this time around, there's a close second (collections) and there's *so much else* I consider it the most important update modders have ever gotten. I will be assuming your approach is like Equestria at War's (total integration of all new features) rather than just doing the bare minimum to get it able to start, ignoring all the cool new features. I have arbitrarily categorized these as **player features** (those the player will directly interact with) or **modder features** (those they will not).

A note on DLC

Every DLC brings both free features and paid features, and NCNS is no different. Unfortunately, information on *which* features are DLC and which are not has been rather difficult to find. I have put my best guess forward, and in some cases it was clear—but at the end of the day, it's PDX's call. Known-to-be-DLC features are marked in the subheadings below as "DLC".

I do *not* consider features that have DLC elements, like the doctrine rework, to be DLC features. They merely have some DLC-locked elements. Only if the entire feature is locked is the feature considered a DLC feature.

Collections: Major Modder Feature

Collections are almost as big a deal as characters were. This time, rather than overhauling how leaders, generals, advisors, and such work, they're a very, very stylish replacement for arrays in many contexts.

In brief, collections allow you to have **self-updating non-ordered arrays**. You can use them in triggers, effects, and more. Like script constants, they require a prefix, in this case **collections:**. For example, **collections:all_countries** instead of just **all_countries**. Either way, they are *astoundingly* powerful, especially as you can feed them into each other, and even mix them with script constants. For example, check out the following four collections (the first two are vanilla, while the final two are mine):

```
controllable_balkan_states = {
  input = constant:state_groups.balkans
  name = COLLECTION_FACTION_BALKAN_STATES
}

states_controlled_by_faction_member_and_core_holder = {
  input = game:scope
  operators = {
    faction_members
    controlled_states
    limit = {
      is_core_of = controller
    }
  }
  name = COLLECTION_STATES_CONTROLLED_BY_FACTION_MEMBER_AND_CORE_HOLDER
}

non_capitulated_majors = {
  input = game:all_countries
  operators = {
    limit = {
      is_major = yes
      has_capitulated = no
    }
  }
  name = COLLECTION_NON_CAPITULATED_MAJORS
}
```

```

}
non_capitulated_majors_with_smaller_navy_than_HIP = {
  input = collection:non_capitulated_majors
  operators = {
    limit = {
      naval_strength_comparison = {
        other = HIP
        ratio < 1.0
        sub_unit_def_weights = { # carriers REALLY scare us, and we don't
really care about small ships
          carrier = 1.45
          battleship = 1.15
          battle_cruiser = 1
          heavy_cruiser = 0.85
          light_cruiser = 0.7
          destroyer = 0.55
          submarine = 0.3
        }
      }
    }
  }
}
name = COLLECTION_NON_CAPITULATED_MAJORS_WITH_SMALLER_NAVY_THAN_HIP
}

```

Collections are defined in `common/collections/`. They can also be defined “anonymously” if you don't need them elsewhere, similar to temp variables.

I have found the following undocumented collection-related triggers: `count_in_collection` and `has_resources_in_collection`.

Doctrines: Major Player Feature

Most doctrines have been completely overhauled, though some traces of the old pattern still remain (as seen in the special forces doctrines). Doctrines have a few core concepts: **folder**, **grand doctrine**, **track**, **milestone**, **subdoctrine**, **mastery**, and **reward**. Everything here is under the `common/doctrines/` path. It's very heavily customizable, and well-documented too!

Folder

The **folder** is the highest-level doctrine element. By default there are three—land, naval, and air—but you can add more. Inside a folder, you choose a **grand doctrine**.

```
land = {  
    allowed = {  
        always = yes  
    }  
    name = "land_doctrine_folder"  
    ledger = army  
    ledger_gfx = GFX_land_doctrine_folder_icon  
    tab_gfx = GFX_landdoctrine_tab_large  
    color_frame = 1  
    sound = ui_doctrine_tab_land  
}
```

Grand Doctrine

Grand doctrines should be familiar: they're things like "fleet in being" or "grand battleplan". They have bonuses at the start when selected, and bonuses for completing a **track**. They cost XP to unlock, usually 100.

Their syntax looks a little weird at first, but you get used to it (the infantry milestone stuff is for achievements):

```
grand_battleplan = {  
    folder = land  
  
    name = GRAND_DOCTRINE_GRAND_BATTLEPLAN  
    description = GRAND_DOCTRINE_GRAND_BATTLEPLAN_DESC  
    icon = GFX_doctrine_grand_battleplan_medium  
    available = {  
        always = yes  
    }  
  
    xp_cost = 100  
    xp_type = army  
  
    ai_will_do = {  
        base = 2  
    }  
}
```

```

tracks = {
    infantry
    combat_support
    armor
    operations
}

# EFFECTS
enable_tactic = tactic_planned_attack

max_planning = 0.1
command_power_gain = 0.25

milestones = {
    {
        #Infantry
        additional_brigade_column_size = 1
        enable_tactic = tactic_elastic_defense
        max_dig_in_factor = 0.05

        effect = {
            if = {
                limit = {
                    NOT = {
                        has_variable = infantry_milestone_var
                    }
                }
                set_variable = { infantry_milestone_var = 1 }
            }
            else = {
                add_to_variable = { infantry_milestone_var = 1 }
            }
        }
    }
    {
        #Artillery/Support
        coordination_bonus = 0.1
        enable_tactic = tactic_overwhelming_fire
    }
}

```

```

    effect = {
        if = {
            limit = {
                NOT = {
                    has_variable = support_milestone_var
                }
            }
            set_variable = { support_milestone_var = 1 }
        }
        else = {
            add_to_variable = { support_milestone_var = 1 }
        }
    }
}
{
    #Armor
    land_reinforce_rate = 0.02
    enable_tactic = tactic_breakthrough
    effect = {
        if = {
            limit = {
                NOT = {
                    has_variable = armor_milestone_var
                }
            }
            set_variable = { armor_milestone_var = 1 }
        }
        else = {
            add_to_variable = { armor_milestone_var = 1 }
        }
    }
}
{
    #Operations
    coordination_bonus = 0.1
    max_command_power = 30

```

```

    effect = {
        if = {
            limit = {
                NOT = {
                    has_variable = operations_milestone_var
                }
            }
            set_variable = { operations_milestone_var = 1 }
        }
        else = {
            add_to_variable = { operations_milestone_var = 1 }
        }
    }
}
}
}
}

```

Track

A **track** holds a **subdoctrine**, and appears very minimal code-wise. Almost everything related to a track is handled by either the **grand doctrine** or the **subdoctrine**. Really, all a **track** does is say what it takes to get **mastery** for that track, and how much you get.

```

combat_support = {
    name = DOCTRINE_TRACK_COMBAT_SUPPORT
    background = "GFX_sup_firepower_bg"
    icon = "GFX_doctrine_milestone_artillery_land"
    icon_frame = "GFX_doctrine_decor_land"
    mastery = {
        multiplier = 10.0
        categories = {
            category_line_artillery
            category_artillery
            category_support_battalions
        }
    }
}
}

```

Milestone

A **milestone** is a reward from completing a **track** with any **subdoctrine**. They are linked to **grand doctrine**, not to the specific **subdoctrine** you chose. They do not have their own subdirectory; they're implicitly handled in the **grand doctrine** directory.

Subdoctrine

A **subdoctrine** is chosen per **track** for some XP (usually 100), with customizable **mastery** requirements. Once you select this **subdoctrine**, you get an immediate bonus, and then begin progressing down it based upon **mastery**. There are many **rewards** along the way, each with its own bonuses.

```
air_cavalry = {
    track = combat_support
    name = SUBDOCTRINE_AIR_CAVALRY
    description = SUBDOCTRINE_AIR_CAVALRY_DESC
    icon = GFX_doctrine_air_cavalry_medium

    xp_cost = 100
    xp_type = army

    available = {
        is_special_project_completed = sp:sp_air_helicopter
    }
    visible = {
        has_dlc = "Gotterdammerung"
    }

    ai_will_do = { base = 0 }

    # EFFECTS
    effect = {
        add_tech_bonus = {
            bonus = 0.5
            uses = 2
            category = helicopter_tech
            name = SUBDOCTRINE_AIR_CAVALRY
        }
    }
}
```



```
rewards = {
    airborne_observers = {
        helicopter_recon = {
            battalion_mult = {
                category = category_artillery
                soft_attack = 0.1
            }
            max_organisation = 10
        }
    }
    service_and_support = {
        category_helicopter_support_companies = {
            max_strength = 0.03
        }
    }
    rescue_liaisons = {
        helicopter_field_hospital = {
            battalion_mult = {
                category = category_all_infantry
                max_strength = 0.05
                default_morale = 0.05
            }
            max_organisation = 5
            casualty_trickleback = 0.05
        }
    }
    transport_assault = {
        helicopter_transport = {
            battalion_mult = {
                category = category_light_infantry
                maximum_speed = 0.2
                breakthrough = 0.05
            }
            breakthrough = 0.05
            max_organisation = 10
        }
    }
    intervention_units = {
        helicopter_brigade = {
```

```

        battalion_mult = {
            category = category_all_infantry
            soft_attack = 0.1
        }
        max_organisation = 10
    }
}
}
}
}

```

Mastery

Mastery is a new resource that is earned by training or fighting with units of a specific kind. More granular than XP, this resource is *only* used in doctrines (currently at least). Certain things can modify its gain, and this is a readily-available buff for old doctrine buffs! See below for an example:

```

add_mastery_bonus = {
    bonus = 0.2
    days = 180
    grand_doctrine = mass_assault
}

```

Reward

A **reward** is the “checkpoint” for a certain **mastery** level inside a **subdoctrine**. It is where most of the buffs come from.

Factions: Major Player DLC Feature

Factions are actually customizable beyond diplo triggers now. They have a *ton* of options available and the documentation is fantastic, so go start by reading that! Note that while you can use the effect `create_faction_from_template` in dlc-neutral contexts, **all advanced faction-related things are DLC-locked**.

Back? Okay, let’s look at an example faction. Looks simple enough, but there’s a lot hidden in here:

```

faction_template_river_coalition = {
    name = river_coalition_faction
    manifest = faction_manifest_ruling_the_rivers
}

```

```

goals = {
    faction_goal_an_infantry_army # short
    faction_goal_a_great_fortress # medium
    faction_goal_defeat_the_empire # long
}

default_rules = {
    call_to_war_rule_faction_leader_offensive_only
    faction_peace_take_claimed_focus
    government_in_exile_allowed
}

icon = GFX_faction_icon_allies
}

```

Rule

Rules are each defined in the `common/rules/` directory, and grouped in `common/rules/groups`. Here's one of them:

```

all_to_war_rule_faction_leader_offensive_only = {
    type = call_to_war_rules
    trigger = {
        OR = {
            is_faction_leader = yes
            has_defensive_war = yes
        }
    }

    visible = {
        has_faction_template = faction_template_river_coalition
    }
    modifier = {
        defensive_war_stability_factor = 0.05
    }
}

```

Goal

And here's how a goal looks. Note they can be short, medium, or long term:

```

faction_goal_defeat_the_empire = {

```

```

name = defeat_the_empire_name
description = defeat_the_empire_desc
category = long_term
group = FOCUS_FILTER_ANNEXATION

allowed = {
    is_griffon_continent_country = yes
    NOT = {
        original_tag = GRI
        original_tag = BRZ
        original_tag = FEA
        original_tag = ANG
        original_tag = KAT
        original_tag = PYT
        original_tag = ROU
        original_tag = YAL
        original_tag = STW
    }
}

completed = {

    GRD = {
        OR = {
            is_in_faction_with = ROOT
            has_capitulated = yes
            exists = no
            is_subject_of = ROOT
        }
    }

}

complete_effect = {
    add_faction_power_projection = 500
    custom_effect_tooltip = generic_skip_one_line_tt
    custom_effect_tooltip = THE_FACTION_LEADER
    add_faction_initiative = 1
    custom_effect_tooltip = generic_skip_one_line_tt
    every_faction_member = {

```

```

        tooltip = INFLUENCE_TOP_THREE
        visible_when_empty = yes
        limit = {
            faction_influence_rank < 4
        }
        add_political_power = 100
    }

}

```

Manifest

This is what a manifest looks like:

```

faction_manifest_the_greatest_navy_in_the_world = {
    name = faction_manifest_the_greatest_navy_in_the_world_name
    description = faction_manifest_the_greatest_navy_in_the_world_desc
    is_manifest = yes
    ratio_progress = {
        total_amount_collection = collection:non_capitulated_majors
        completed_amount_collection =
collection:non_capitulated_majors_with_smaller_navy_than_HIP

    scale = {
        navy_fuel_consumption_factor = -0.25
        fuel_gain_factor = 0.25
    }
    progress_sections = {
        manifest_fulfilled = {
            min = @manifest_fulfilled_value
            modifier = {
                production_lack_of_resource_penalty_factor = -0.2
            }
        }
    }
}
}

```

So as you can see, there's a lot going on! Check around the documentation and the examples to get a feel for the system. There are also **member upgrades** for things like manpower sharing, but these are a little less common to be modded (in my experience anyway).

AI Faction Theatres: Modder Feature

With the new faction upgrades, we also have the ability to create theatres—and the ability to tell the AI which theatres to build. Here's an example:

```
alicorn_sunset_north = { # northern front down to crystal ponies
    name = theater_alicorn_sunset_north
    regions = {
        84 # Featherfall Dale
        73 # Yaket Range
        74 # Stonefields
        71 # Crystal Heart
        72 # Crystal Range
        65 # Severyana
        102 # Central Yakistan
        103 # Greater Yakistan Valley
        12 # Frozen North
        191 # Bronze Mountains
        121 # Bright Valleys
    }

    preferred_countries = {
        PLB
    }

    cancel = {
        NOT = { has_war_with = EQS }
    }

    ai_will_do = {
        base = 0
        modifier = {
            add = 5
            OR = {
                original_tag = CHN
            }
        }
    }
}
```

```

        is_in_faction_with = CHN
    }
    has_war_with = EQS
}
modifier = {
    add = 1
    is_in_faction_with = CHN
    has_war_with = EQS
}
}
}

```

AI Navy: Modder Feature

AI Navy can finally be modded extensively! Beyond just equipment, we can tell different tags what kinds of taskforces and fleets to build, and when. They also can be told what the role should be. They're well-documented, so go read the documentation!

Back? Excellent. There are four big ideas here: **fleets**, **taskforces**, **goals** and **objectives**. Goals are high-level things like protecting trade routes, while objectives are low-level things like protecting a specific trade route. Fleets are the groupings of taskforces, while taskforces attempt to carry out specific missions. Objectives are developed by the AI itself and thus **hardcoded**. The other three, however, are moddable!

Fleets

Here's a sample fleet file:

```

HIP_dominance_fleet_1 = {
    required_taskforces = {
        HIP_StrikeForce_1 = 1
        HIP_PatrolDominanceForce_1 = 3
        HIP_PatrolReconForce_1 = 3
    }
    optional_taskforces = {
        HIP_StrikeForce_1 = 1
        HIP_PatrolDominanceForce_1 = 3
        HIP_PatrolReconForce_1 = 3
    }
}
}

```

```

HIP_raiding_fleet_1 = {
    required_taskforces = {
        HIP_ConvoyRaiding_1 = 4
    }
    optional_taskforces = {
        HIP_ConvoyRaiding_1 = 2
    }
}

HIP_escort_fleet_1 = {
    required_taskforces = {
        HIP_ConvoyEscort_1 = 2
    }
    optional_taskforces = {
        HIP_ConvoyEscort_1 = 4
    }
}

HIP_minelaying_fleet_4 = {
    required_taskforces = {
        HIP_MineLaying_1 = 1
    }
}

```

Each fleet contains **taskforces**. Some have only one (like the minelaying), while some have many like the dominance. Let's look at the construction of a HIP dominance taskforce:

Taskforces

```

HIP_StrikeForce_1 = {
    allowed = {
        original_tag = HIP
    }
    ai_will_do = {
        factor = 1
    }
    mission = { naval_strike }
    min_composition = {
        destroyer = {

```



```

        amount = 8
    }
}

optimal_composition = {
    carrier = {
        amount = 2
    }
    battleship = {
        amount = 4
    }
    heavy_cruiser = {
        amount = 4
    }
    light_cruiser = {
        amount = 4
    }
    destroyer = {
        amount = 36
    }
}

}

HIP_PatrolReconForce_1 = {
    allowed = {
        original_tag = HIP
    }
    ai_will_do = {
        factor = 5
    }
    mission = { naval_patrol }
    min_composition = {
        light_cruiser = {
            amount = 2
        }
    }
}
    optimal_composition = {
        light_cruiser = {

```

```

        amount = 3
    }
}

HIP_PatrolDominanceForce_1 = {
    allowed = {
        original_tag = HIP
    }
    ai_will_do = {
        factor = 5
    }
    mission = { naval_patrol }
    min_composition = {
        destroyer = {
            amount = 4
        }
    }

    optimal_composition = {
        heavy_cruiser = {
            amount = 3
        }
        destroyer = {
            amount = 12
        }
    }
}

```

As you can see, fleets list actual ships in amounts and what they are supposed to do! This is then fed into the goal-objective system as follows (for HIP):

Goals

```

HIP_naval_invasion_support = {
    objective_type = naval_invasion_support

    available_for = {
        HIP
    }
}

```

```
    min_priority = 10
    max_priority = 20
}

HIP_mine_sweeping = {
    objective_type = mines_sweeping

    available_for = {
        HIP
    }

    min_priority = 3
    max_priority = 8
}

HIP_invasion_defense = {
    objective_type = naval_invasion_defense

    available_for = {
        HIP
    }

    min_priority = 10
    max_priority = 18
}

HIP_coast_defense = {
    objective_type = coast_defense

    available_for = {
        HIP
    }

    min_priority = 1
    max_priority = 10
}

HIP_convoy_protection = {
```

```
    objective_type = convoy_protection

    available_for = {
        HIP
    }

    min_priority = 1
    max_priority = 5
}

HIP_convoy_raiding = {
    objective_type = convoy_raiding

    available_for = {
        HIP
    }

    min_priority = 3
    max_priority = 7
}

HIP_naval_dominance = {
    objective_type = naval_dominance

    available_for = {
        HIP
    }

    min_priority = 1
    max_priority = 13
}

HIP_mine_laying = {
    objective_type = mines_planting

    available_for = {
        HIP
    }
}
```

```

    min_priority = 1
    max_priority = 5
}

HIP_training = {
    objective_type = training

    available_for = {
        HIP
    }

    min_priority = 10
    max_priority = 20
}

```

This lets the AI have its behavior “clamped”, so for example HIP never goes all in on minelaying and always worked hard on coastal defense.

Focus Dynamic Lengths & Overlay: Modder Feature

After so many years, it’s here! For you, this means you can make a focus’s length whatever you want, and have it updated within the tree. Here’s how you would reduce a focus by 70 days:

```

    reduce_focus_completion_cost = {
        focus = PHI_national_defense_act
        cost = 70
    }

```

Focuses also support *overlays*, things that appear atop them similar to titlebar styles or the flags from joint focuses (but can be combined with those). They simply add an argument to the focus block that looks like this:

```

overlay = GFX_focus_fast_overlay_generic_clock

```

Energy & Coal: Player Feature

A new resource—**coal**—was added, along with the idea of “powered” factories. You also have modifiers that only apply when fully powered, though these are fairly rare.

```

production_speed_buildings_powered_factor, industrial_capacity_factory_powered,
industrial_capacity_dockyard_powered
factory_energy_consumption, local_factory_energy_consumption,
local_resources_<resource>_factor

```

Note that **the more mobilized your economy, the more energy it takes**. Currently, vanilla does not have techs for increasing energy gain or adding energy directly, nor the modifiers necessary to mod them in. Something to hope for!

Furthermore, note that a new NAI define was added:

```
NDefines.NAI.MIN_FACTORIES_TO_WANT_TO_IMPORT = { -- minimum number of civilian
factories the AI must have to consider importing a resource - per strategic
resource. Default 0, array -should- be updated with new resources, or if the
order changes.
    10, -- oil -- EAW increased because of outdated societies
    0, -- aluminium
    0, -- rubber
    0, -- tungsten
    0, -- steel
    0, -- chromium
    10, -- coal
    0, -- crystals
}
```

This **must** be updated with any new resources your mod has, or the AI won't understand how it works!

Naval Dominance: Player Feature

Naval supremacy is gone. Now it's naval **dominance**. The entire system works differently, with naval terrain affecting the amount required to dominate, different missions interacting with the naval dominance in different ways, and ships having hugely varying amounts of naval dominance (but visible in the designer!). This is a **big** change, so dig around and see what there is! In particular, note that `common/00_terrain.txt` **requires dominance values to work**.

Ranger Special Forces: Player Feature

At long last, we have special forces built for forests and jungles, and the attendant special forces doctrines if you also have AAT.

Strategic Locations: Player Feature

Strategic Locations are a new feature that allow us to customize the amount of maximum province or state buildings in a specific state or province. They come in two categories: province

and state. Wonderfully, these are defined the same way, just with a different effect, as seen in Okinawa's state file (under the history block):

```
    strategic_province_location = {
        defensible_coastline = 10124
    }
    strategic_state_location = {
        favorable_approach = 1123
    }
    strategic_province_location = {
        defensive_chokepoint = 11932
    }
```

They're defined in `common/strategic_locations/`, and can take any building type with a cap. They take the same effect to add during the game, though note it must be in state scope! For an example, see:

```
defensible_coastline = {
    coastal_bunker = 2
}
favorable_approach = {
    air_base = 2
}
```

Expanded Player Features

Here are the updated and new things that the player will care about that are not whole new categories.

New DLC Techs, SPs & Modules

Navies are no longer purely combat. **Repair ships** and **support ships** offer powerful buffs to your navy, including "saving" ships that would otherwise be sunk! They're handled through two new equipment types, and a static modifier called in the code:

```
# This is scaled based on support ratio
naval_general_support = {
    max_organisation = 0.2
    navy_max_range_factor = 0.2
    naval_morale_factor = 0.2
}
# This is scaled based on support ratio
naval_repair_support = {
```

```

    naval_critical_effect_factor = -0.25
    naval_ship_recovery_chance = 0.1
    naval_attrition = -0.075
}

```

There are new toys like **sonar 3 and 4** and **seaplane tenders** (no more invisible subs). We've also got the *excellent* additions of submarine carriers and escort carriers as special projects.

Rework to Carrier Overstacking

Finally! The defines for the old-style carrier method were removed. Now, it's based around the number of ships in service, so the larger the battle, the more planes can operate effectively:

```

    NAVAL_COMBAT_PLANE_MIN_STACKING_PENALTY = 80,                -- How
many planes flying in a naval combat before penalties are introduced
    NAVAL_COMBAT_PLANE_STACKING_PENALTY_EFFECT = 0.025,          -- Each plane
above the optimal amount decreases the amount of airplanes being able to takeoff
by such %. Subject to diminishing returns

    SHIP_SILHOUETTE_VALUE_PLANES_CAPITAL = 10,                  -- For
dynamic plane efficacy, silhouette value (nominally in planes, but very abstract)
    SHIP_SILHOUETTE_VALUE_PLANES_SCREEN = 5,                    -- As Above.
This one would be nice to split by type, but that's problematic.
    SHIP_SILHOUETTE_VALUE_PLANES_CARRIER = 16,                 -- As Above
    SHIP_SILHOUETTE_VALUE_PLANES_SUPPORT = 3,                   -- As Above
    SHIP_SILHOUETTE_VALUE_PLANES_CONVOY = 4,                    -- As Above
    SHIP_SILHOUETTE_VALUE_PLANES_SUBMARINE = 7,                 -- As Above

    SCREEN_CAP_REDUCTION_FACTOR = 0.02,                          -- Reduces screen
silhouette weight if there are caps present, screenval * 1/(1+caps*weight)

```

So as you can see, against a “small” taskforce of 1 carrier, 1 battleship, 1 heavy cruiser, 2 light cruisers, and 8 destroyers, the number of planes able to fly without penalty is $80+16+10+10+(5*2+5*8)*(1/(1+3*0.02))=163.2$.

World Tension Overhaul

World tension has gotten some significant changes in the defines. Before, puppeting gave tension for example. Now it doesn't, at all. TENSION_STATE_VALUE used to be divided by 20 but no longer is, and TENSION_PEACE_FACTOR is now reduced by a new define: TENSION_PEACE_THREAT_FACTOR. No-CB wargoals are more brutal, annexing a core now

reduces world tension, and forcing government now *reduces* world tension. Taking capital ships no longer generates tension, nor do war reparations, resource rights, dismantling industry, or capitulation.

Carrier Detection Based on Planes

This is a really cool one. Now, the more planes you have on a carrier, the better it is at hunting down enemy ships. Different carrier modules support different amounts, so it's heavily moddable!

```
add_stats = {  
    carrier_size = 2  
    build_cost_ic = 1500  
    carrier_surface_detection = 5  
    carrier_sub_detection = 2  
}
```

New Buildings

We have **naval supply hubs** and **naval HQs**. Naval supply hubs reduce the supply consumption of fleets, and naval HQs let you assign admirals to give wide-ranging boosts (DLC-only). Interestingly, these new buildings require at least a level 8 port in the province—something I didn't think was possible before. Here's the syntax:

```
building_prerequisite = {  
    building_dependency = {  
        building = naval_base  
        level = 8  
    }  
}
```

State Category Tweaks

Now state categories can support limits on the maximum building level with them—things like

```
buildings_max_level = {  
    naval_base = 6  
    air_base = 4  
}
```

Currently, these are only capped on islands. Additionally, there's a new vanilla state category—**large island**. This has 3 building slots and a naval base/air base cap.

Miscellaneous Balance Changes

There are WAY too many to list here, so poke around the files. For example, snow and cold are more brutal now, and naval invasions work entirely differently. Medium battery techs don't exist anymore, and the modules got rolled into light and heavy batteries.

Expanded Modder Features

Here are the updated and new things that the player likely won't notice that are not whole new categories.

New Bookmark Features

Bookmarks support some new arguments now. They have the following two options under a country:

```
new_content = yes
override_leader_portrait = JAP_emperor_hirohito
```

So you can have the icon pulse to notify the player of what's new, as well as show a different portrait than is actually there!

Clickable Focus Inlay Window Buttons and Inlay Offset

The GUIfication of vanilla continues! Now focus inlay windows support buttons that have effects and triggers. Example:

```
army_faction_stamp_of_approval = {
    available = {
        has_political_power > 74
        JAP_army_faction_is_not_dominant = yes
        IF = {
            limit = {
                NOT = { has_country_flag =
JAP_interfered_in_imperial_influence_flag }
            }
            NOT = { has_country_flag =
JAP_interfered_in_imperial_influence_flag }
        }
        ELSE_IF = {
            limit = {
                has_country_flag =
JAP_interfered_in_imperial_influence_flag
            }
        }
    }
}
```

```

        custom_trigger_tooltip = {
            tooltip =
JAP_imperial_influence_interference_reset_days_tt
            NOT = { has_country_flag =
JAP_interfered_in_imperial_influence_flag }
        }
    }
    NOT = {
        has_country_flag = JAP_blocked_from_favoring_the_army_flag
    }
}
click_effect = {
    IF = {
        limit = {
            JAP_army_faction_is_tier_1 = yes
        }
        add_political_power = -200
    }
    ELSE = {
        add_political_power = -75
    }
}

custom_effect_tooltip = generic_skip_one_line_tt

JAP_big_army_faction_gain = yes
set_country_flag = {
    flag = JAP_interfered_in_imperial_influence_flag
    value = 1
    days = 180
}
}
}

```

We also now can have offsets to inlays, or rather override_position:

```

override_position = {
    x = 4400
    y = 700
    trigger = {
        has_game_rule = {
            rule = obsolete_focus_branches_visibility

```

```

        option = HIDE
    }
    has_completed_focus = JAP_sea_purge_the_kodoha_faction
}

override_position = {
    x = 6000
    y = 700
    trigger = {
        has_game_rule = {
            rule = obsolete_focus_branches_visibility
            option = HIDE
        }
        has_completed_focus = JAP_revere_the_emperor_destroy_the_traitors
    }
}

```

This goes in the common/national_focus/ file, within the inlay itself.

State and Country Group Script Constants

We've got new script constants for countries and states, as well as a new trigger using it in the following style:

```

any_country_with_original_tag_of = {
    tooltip = TRIGGER_ANY_COUNTRY_STARTS
    target = constant:country_groups.literally_china
    OR = {
        has_completed_focus = CHI_ledo_road
        has_completed_focus = CHI_sea_ledo_road
    }
}

```

Here are how the script constants look:

```

country_groups = {
    schema = {
        any_key = yes
        array = country
    }
    north_zebrica = {
        HIP
    }
}

```

```
    WIN
    BAT
    CTH
    ZAR
    WAR
    TBK
    ZUM
    CVA
    TRO
    HIE
    CDR
    CGI
    BKH
  }
}
```

A similar concept exists for state_groups:

```
state_groups = {
  schema = {
    any_key = yes
    array = state
  }

  # Balkans including Continental Greece + Turkey and all of 1936 Romania (But
not Hungary)
  balkans = {
    102
    853
    103
    109
    104
    804
    45
    764
    107
    108
    803
    105
    802
    106
  }
}
```

```
970
934
44
805
163
852
736
48
212
801
211
77
971
184
731
185
47
186
46
81
82
83
84
76
79
766
78
80
341
797
}
}
```

Force Linking Impassable States

No more will we have to deal with impassables just flipflopping like crazy. An impassable state now supports the argument `force_link_ownership_to = [STATE_ID]`, which makes its ownership equivalent to the owner of that state. This is set in the history file, usually right after the `impassable = yes`.

Miscellaneous New Triggers

has_resources_in_collection	- country scope, collection stuff
is_on_same_continent_as	- country, state scope
faction_power_projection	- country scope
has_offensive_war_without_friend	- country scope
can_assign_supportive_scientist_to_faction	- country scope
faction_upgrade_level	- country scope
faction_influence_score	- country scope
faction_influence_ratio	- country scope
has_manpower_to_become_leader	- country scope
has_industry_to_become_leader	- country scope
count_in_collection	- country scope, collection stuff
collection_contains	- any scope, collection stuff
has_doctrine	- country scope
has_completed_subdoctrine	- country scope
has_subdoctrine_in_track	- country scope
has_mastery	- country scope
has_completed_track	- country scope
has_tactic	- country scope
has_enemy_naval_control	- country scope
has_naval_control	- country scope
any_state_of	- any scope
any_collection_element	- any scope, collection stuff
all_collection_elements	- any scope, collection stuff
all_country_of	- any scope
all_claimant	- state scope
any_country_of	- any scope
any_country_with_original_tag_of	- any scope
any_other_country_of	- any scope
any_other_country_with_original_tag_of	- any scope
has_faction_template	- country scope
faction_manifest_fulfillment	- country scope
collection_size	- any scope, collection stuff
faction_influence_rank	- country scope
has_completed_faction_goal	- country scope
has_faction_goal	- country scope
faction_goal_fulfillment	- country scope
is_leading_army_in_province	- character scope
has_active_rule	- country scope
has_faction_research_unlocked	- country scope
has_faction_military_unlocked	- country scope
has_enough_influence_for_leadership	- country scope
compare_ideology_with_faction	- country scope

Miscellaneous New Modifiers

army_claim_attack_factor
naval_critical_score_chance_factor_against
naval_supply_consumption_factor
navy_capital_ship_attack_factor_against
navy_screen_attack_factor_against
navy_submarine_attack_factor_against
planning_decay_rate_factor
air_invasion_division_cap
air_invasion_preparation
army_claim_defence_factor
army_experience_from_volunteers
army_retreat_speed_factor
modifier_army_sub_unit_category_rocket_artillery_attack_factor
rangers_special_forces_contribution_factor
autonomy_manpower_share_from_subjects
resources_to_overlord_factor
<Doctrine>_doctrine_cost_factor
<Doctrine>_mastery_gain_factor
<Doctrine>_track_mastery_gain_factor
<IdeaCategory>_category_type_cost_factor
<IdeaGroup>_cost_factor
<ModifierStat>_factor # "naval_general_support", "naval_repair_support"
<ModifierStatValue>_value_factor # "naval_general_support",
"naval_repair_support"
<Technology>_cost_factor
air_invasion_plan_cap
air_invasion_prep_days
annex_subject_cost_factor
automatic_grant_medal_chance
experience_gain_<Unit>_mission_factor
factory_energy_consumption
industrial_capacity_dockyard_powered
industrial_capacity_factory_powered
local_resources_<Resource>_factor
naval_invasion_plan_cap
naval_invasion_prep_days
out_of_power_impact_factor
production_cost_<Building>_factor
production_speed_buildings_powered_factor
repair_speed_<Building>_factor
shore_bombardment_collateral_damage_factor
special_forces_doctrine_cost_factor
unit_medal_effectiveness
navy_capital_ship_defence_factor_against

navy_screen_defence_factor_against
navy_submarine_defence_factor_against
faction_influence_contribution_factor
faction_influence_war_score_factor
faction_subject_contribution_gain
intelligence_operation_speed
amphibious_invasion_against
experience_gain_<Unit>_combat_factor
experience_gain_<Unit>_training_factor
headquarters_experience_gain_factor
invasion_preparation_against
max_organisation
naval_hit_chance_against
naval_invasion_division_cap
naval_mission_xp_factor
naval_ship_recovery_chance
naval_ship_recovery_chance_factor
spotting_chance_against
local_factory_energy_consumption
can_guarantee_other_ideologies
<ideology>_drift_from_guarantees

Deprecated Effects

create_faction - use `create\_faction\_from\_template`

Miscellaneous New Effects

reduce_focus_completion_cost	- country scope
strategic_province_location	- state scope
strategic_state_location	- state scope
select_tech_tree_icon_origin	- country scope
add_faction_power_projection	- country scope
add_state_resistance_compliance_modifier	- any scope?
remove_state_resistance_compliance_modifier	- any scope?
set_faction_member_upgrade_min	- country scope
set_faction_upgrade	- country scope
every_faction_member	- country scope
add_faction_influence_score	- country scope
add_faction_goal_slot	- country scope
add_faction_influence_ratio	- country scope
set_grand_doctrine	- country scope
set_sub_doctrine	- country scope

add_mastery	- country scope
add_daily_mastery	- country scope
unlock_tactic	- country scope
add_mastery_bonus	- country scope
create_faction_from_template	- country scope
set_faction_rule	- country scope
set_faction_manifest	- country scope
add_faction_goal	- country scope
remove_faction_goal	- country scope
add_faction_initiative	- country scope
every_collection_element	- any scope
set_faction_research_unlocked	- any scope
set_faction_military_unlocked	- any scope

New Variables

```
### total amount of resources extracted by the world
global_resource_extracted_total

### total amount of specific resource extracted by the world, resource type is
defined in target global_resource_extracted@steel
global_resource_extracted@<resource>

### Checks influence ratio of current country in the faction
faction_influence_ratio

### total amount of resources extracted by the country's faction
faction_resource_extracted_total

### total amount of specific resource extracted by the country's faction,
resource type is defined in target faction_resource_extracted@steel
faction_resource_extracted@<resource>

### Checks power value of current country's faction projection
faction_power_projection

### Checks influence value of current country in the faction
faction_influence_score

### Checks manifest fulfillment value of current country's faction manifest
faction_manifest_fulfillment

### Checks influence rank in the faction of the current country
faction_influence_rank

### Total constructions of naval_headquarter
total_constructed_naval_headquarter

### Total constructions of naval_supply_hub
total_constructed_naval_supply_hub
```

```
### Total produced equipment of typesupport_ship  
total_equipment_produced_support_ship
```

最后，向大家推荐一下一个代码制作交流群和美工交流群，欢迎各位对代码（美工）感兴趣的萌新和代码（美工）功成的大佬加入：

霜泽图书馆 群号：378525932 霜泽美术馆 群号：976242794

白雪老师的完美教室：768450264

并且向各位附上代码图书馆，内包含诸多代码文字教程和方便制作的软件和工具：

秋起图书馆 GitHub 地址：

<https://github.com/jingzhouzhidi/QIUQI-LIBRARY.git>

并且，为了各位的制作上的方便，我们急需翻译和审核进行钢铁雄心 4wiki 的汉化工作，因而在此发布招新的地址，感谢支持：

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群号：996535362

如本篇翻译出现任何问题：

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